

ARM-1 Microarchitecture: From Hardwired FSM to Discrete Silicon Silicon Realization

Project:

Computer Architecture Bonus Project
(Multi-cycle 8-bit processor design,
simulation, discrete fabrication)

Institution:

Zewail City University of Science and
Technology, CIE Dept.

Team:

Authored by Team 1 (Sophomore Cohort) -
Moaz Elabd, Anas Refaei, Omar Ihab Fared,
Ahmed Hany, et al.

Supervision:

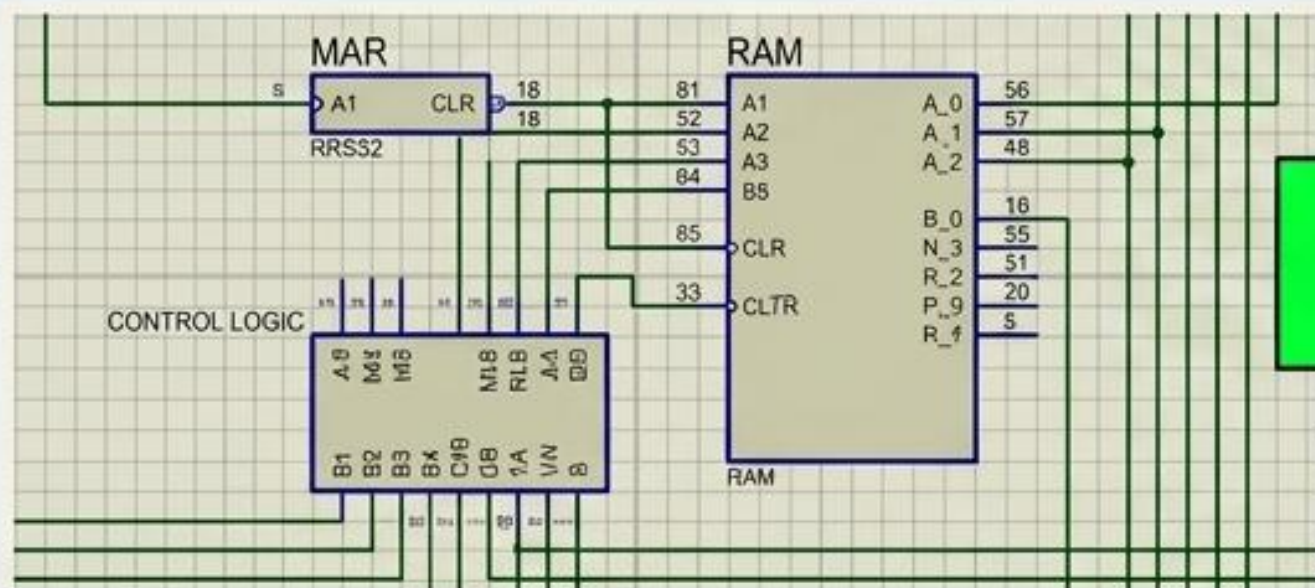
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(Academic Year 2025/2026)

Eng. George S. Maximous.

Eng. Zaynap Ahmed

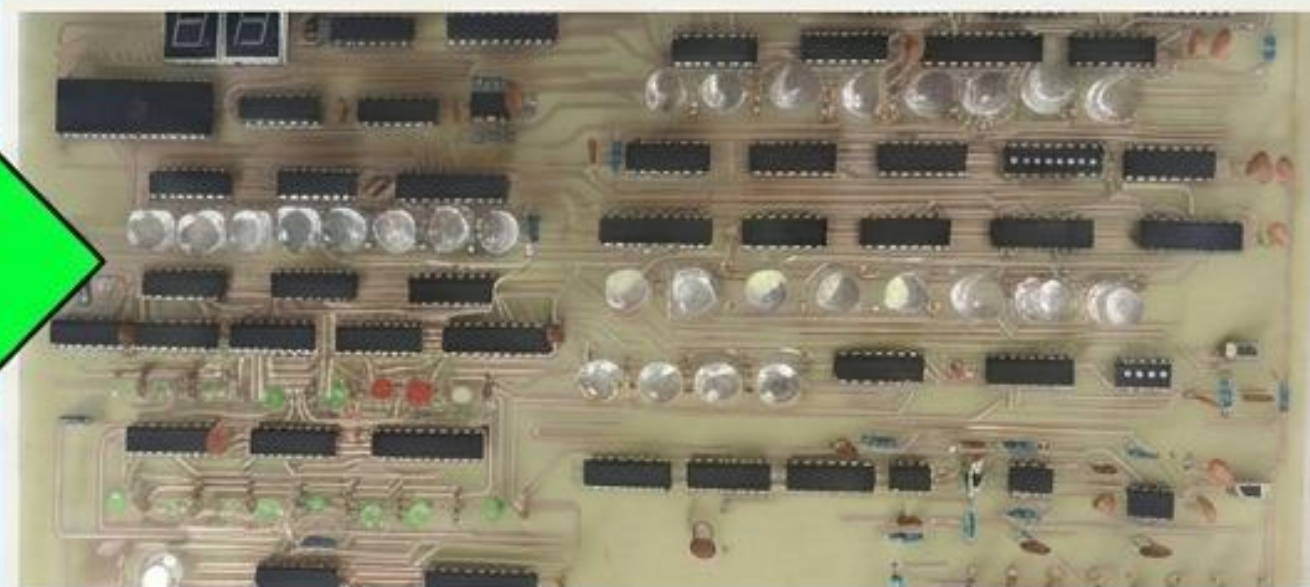
Bridging the Digital and Physical Domains

DIGITAL DOMAIN



TRANSLATION
TO SILICON

PHYSICAL DOMAIN



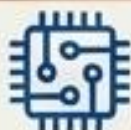
Pedagogical Significance

Models a classical accumulator-based architecture. Strips away pipeline complexities to expose raw, deterministic execution.



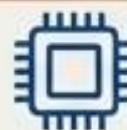
Execution Paradigm

Strict, atomic 3-cycle state machine (Fetch, Decode, Execute). Guarantees 100% deterministic instruction throughput.



Digital Simulation

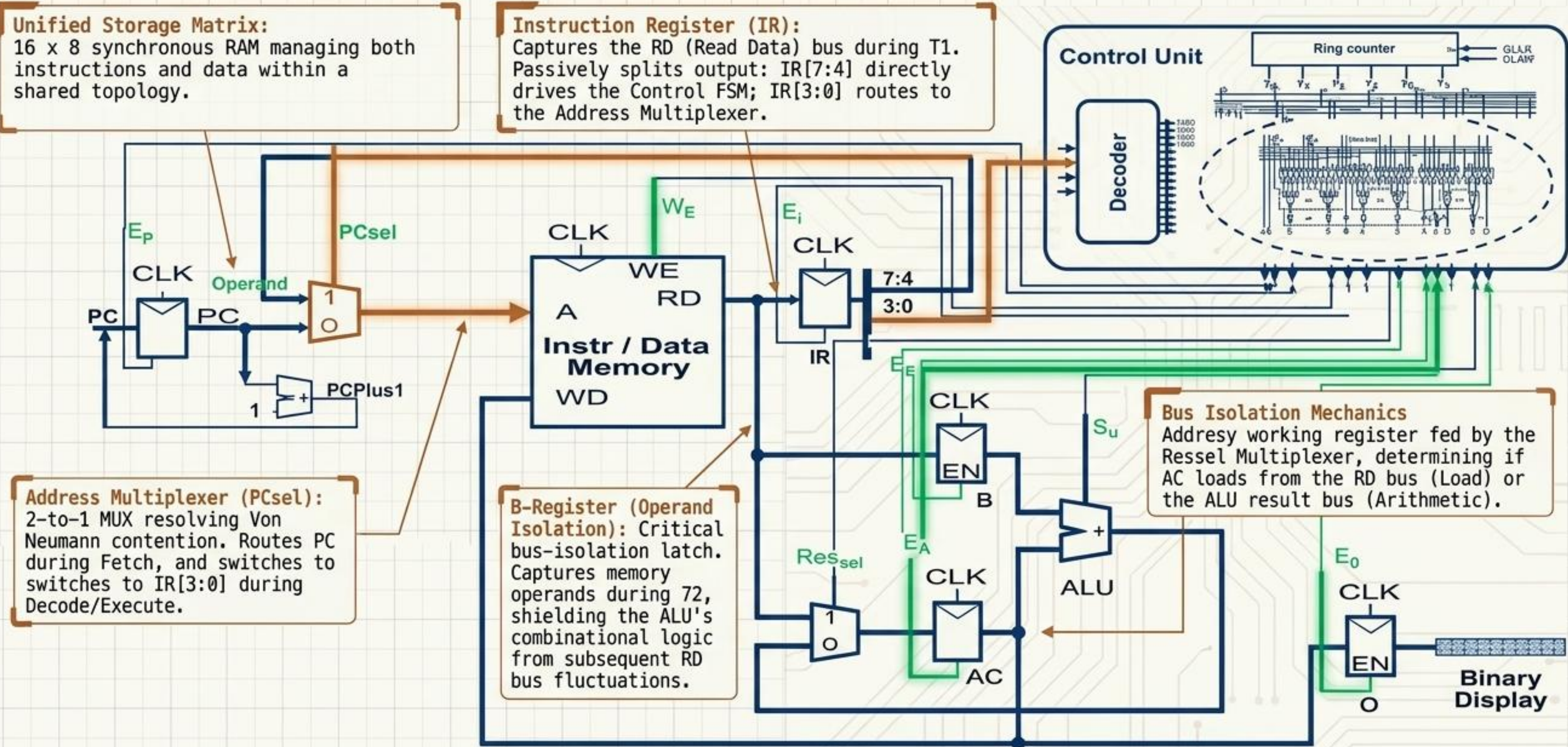
Initial gate-level Logisim verification transitioned to Proteus power-plane mapping.



Physical Realization

Pure, hardwired digital logic. Built entirely from standard 74-series TTL logic ICs. No microcontrollers utilized.

Macro Architecture: The Von Neumann Bottleneck & Shared Bus Routing



Datapath Integration & Execution Registers

Address Multiplexing (PCsel)

Acts as the memory bus traffic controller. Routes the Program Counter (PC) to memory during the Fetch state. Switches to route the lower nibble of the Instruction Register (IR[3:0]) during Decode and Execute states.

Instruction Latching (IR)

Passively captures 8-bit Read Data word during T1.

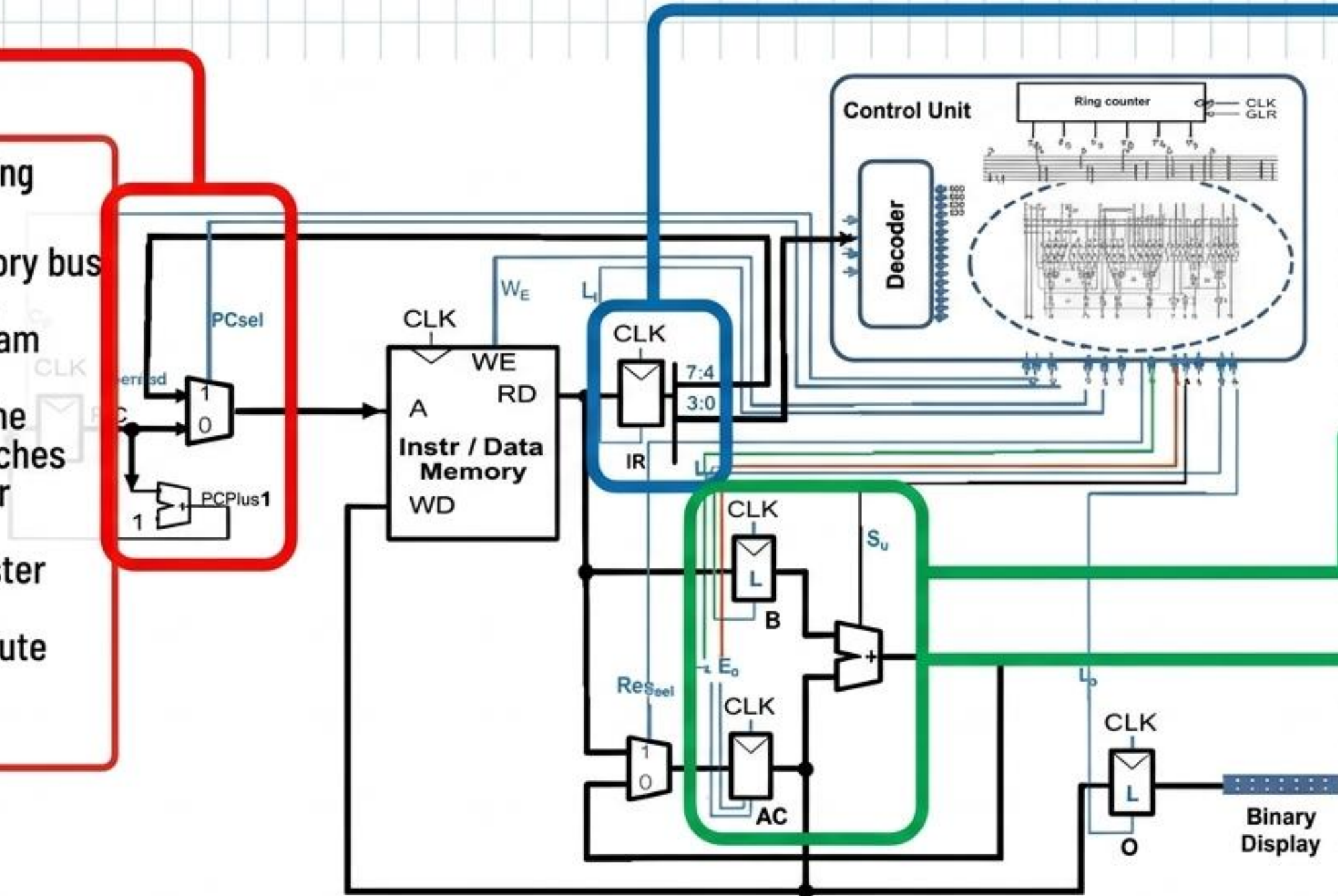
Splits signal: upper 4 bits to hardware Decoder, lower 4 bits to address multiplexer.

ALU & Accumulator (AC)

Primary computation hub. Computes addition ($S_u=0$) or Two's Complement subtraction ($S_u=1$).

Operand Isolation (B-Register)

The critical isolation buffer. Latches the memory operand during T2. Holds ALU input perfectly stable, preventing mid-computation corruption if the shared RD bus fluctuates.



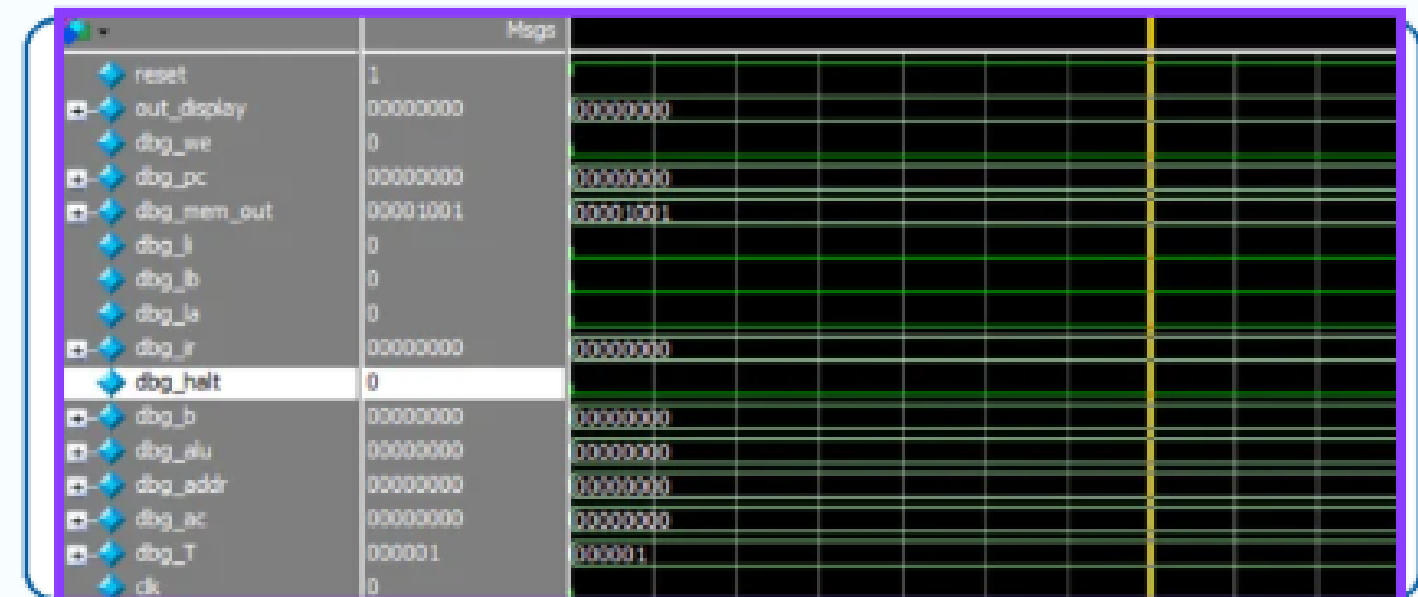
Project 1 Implementation: Testing

Verification of ARM I Processor using ModelSim

Testing Steps

- Test program was loaded inside memory.
- Each module was tested using a separate testbench.
- The full processor was tested using tb_ARM_I_System.
- ModelSim waveforms were used to verify execution.
- Checked PC, IR, AC, B, ALU result, memory output, and control signals.

ModelSim Verification



Individual Testbenches

ALU, Memory,
Register, MUX,
Control Unit

Full System Testbench

tb_ARM_I_System

Expected Output

Program result
verified

Goal: prove that the ARM I datapath, control signals, and output values work correctly.

PROJECT1 SIMULATION RESULT

PROCESSOR EXECUTION SHOWING STEP-BY-STEP CHANGES DURING PROGRAM EXECUTION

T=0	PC=0	IR=0	AC=0	B=0	ALU=0	MEM_OUT=9	ADDR=0	T=000001	LB=0	LA=0	LI=0	WE=0	HALT=0	OUT=0
T=20000	PC=0	IR=0	AC=0	B=0	ALU=0	MEM_OUT=9	ADDR=0	T=000010	LB=0	LA=0	LI=0	WE=0	HALT=0	OUT=0
T=25000	PC=1	IR=0	AC=0	B=0	ALU=0	MEM_OUT=42	ADDR=1	T=000010	LB=0	LA=0	LI=0	WE=0	HALT=0	OUT=0
T=30000	PC=1	IR=0	AC=0	B=0	ALU=0	MEM_OUT=42	ADDR=1	T=000100	LB=0	LA=0	LI=0	WE=0	HALT=0	OUT=0
T=40000	PC=1	IR=0	AC=0	B=0	ALU=0	MEM_OUT=42	ADDR=1	T=001000	LB=0	LA=0	LI=0	WE=0	HALT=0	OUT=0
T=50000	PC=1	IR=0	AC=0	B=0	ALU=0	MEM_OUT=42	ADDR=1	T=010000	LB=0	LA=0	LI=0	WE=0	HALT=0	OUT=0
T=60000	PC=1	IR=0	AC=0	B=0	ALU=0	MEM_OUT=42	ADDR=1	T=100000	LB=0	LA=0	LI=0	WE=0	HALT=0	OUT=0
T=70000	PC=1	IR=0	AC=0	B=0	ALU=0	MEM_OUT=42	ADDR=1	T=000001	LB=0	LA=0	LI=1	WE=0	HALT=0	OUT=0
T=75000	PC=1	IR=2a	AC=0	B=0	ALU=0	MEM_OUT=42	ADDR=1	T=000001	LB=0	LA=0	LI=1	WE=0	HALT=0	OUT=0
T=80000	PC=1	IR=2a	AC=0	B=0	ALU=0	MEM_OUT=42	ADDR=1	T=000010	LB=0	LA=0	LI=0	WE=0	HALT=0	OUT=0
T=85000	PC=2	IR=2a	AC=0	B=0	ALU=0	MEM_OUT=59	ADDR=2	T=000010	LB=0	LA=0	LI=0	WE=0	HALT=0	OUT=0
T=90000	PC=2	IR=2a	AC=0	B=0	ALU=0	MEM_OUT=3	ADDR=a	T=000100	LB=0	LA=0	LI=0	WE=0	HALT=0	OUT=0
T=100000	PC=2	IR=2a	AC=0	B=0	ALU=0	MEM_OUT=3	ADDR=a	T=001000	LB=1	LA=0	LI=0	WE=0	HALT=0	OUT=0
T=105000	PC=2	IR=2a	AC=0	B=3	ALU=3	MEM_OUT=3	ADDR=a	T=001000	LB=1	LA=0	LI=0	WE=0	HALT=0	OUT=0
T=110000	PC=2	IR=2a	AC=0	B=3	ALU=3	MEM_OUT=3	ADDR=a	T=010000	LB=0	LA=1	LI=0	WE=0	HALT=0	OUT=0
T=115000	PC=2	IR=2a	AC=3	B=3	ALU=6	MEM_OUT=3	ADDR=a	T=010000	LB=0	LA=1	LI=0	WE=0	HALT=0	OUT=0
T=120000	PC=2	IR=2a	AC=3	B=3	ALU=6	MEM_OUT=59	ADDR=3	T=100000	LB=0	LA=0	LI=0	WE=0	HALT=0	OUT=0
T=130000	PC=2	IR=2a	AC=3	B=3	ALU=6	MEM_OUT=59	ADDR=2	T=000001	LB=0	LA=0	LI=1	WE=0	HALT=0	OUT=0
T=135000	PC=2	IR=3b	AC=3	B=3	ALU=6	MEM_OUT=59	ADDR=2	T=000001	LB=0	LA=0	LI=1	WE=0	HALT=0	OUT=0
T=140000	PC=2	IR=3b	AC=3	B=3	ALU=6	MEM_OUT=59	ADDR=2	T=000010	LB=0	LA=0	LI=0	WE=0	HALT=0	OUT=0
T=145000	PC=3	IR=3b	AC=3	B=3	ALU=6	MEM_OUT=64	ADDR=3	T=000010	LB=0	LA=0	LI=0	WE=0	HALT=0	OUT=0
T=150000	PC=3	IR=3b	AC=3	B=3	ALU=6	MEM_OUT=0	ADDR=b	T=000100	LB=0	LA=0	LI=0	WE=0	HALT=0	OUT=0
T=160000	PC=3	IR=3b	AC=3	B=3	ALU=6	MEM_OUT=64	ADDR=3	T=001000	LB=1	LA=0	LI=0	WE=0	HALT=0	OUT=0
T=165000	PC=3	IR=3b	AC=3	B=64	ALU=67	MEM_OUT=64	ADDR=3	T=001000	LB=1	LA=0	LI=0	WE=0	HALT=0	OUT=0
T=170000	PC=3	IR=3b	AC=3	B=64	ALU=195	MEM_OUT=64	ADDR=3	T=010000	LB=0	LA=1	LI=0	WE=0	HALT=0	OUT=0
T=175000	PC=3	IR=3b	AC=195	B=64	ALU=131	MEM_OUT=64	ADDR=3	T=010000	LB=0	LA=1	LI=0	WE=0	HALT=0	OUT=0
T=180000	PC=3	IR=3b	AC=195	B=64	ALU=3	MEM_OUT=64	ADDR=3	T=100000	LB=0	LA=0	LI=0	WE=0	HALT=0	OUT=0
T=190000	PC=3	IR=3b	AC=195	B=64	ALU=3	MEM_OUT=64	ADDR=3	T=000001	LB=0	LA=0	LI=1	WE=0	HALT=0	OUT=0
T=195000	PC=3	IR=40	AC=195	B=64	ALU=3	MEM_OUT=64	ADDR=3	T=000001	LB=0	LA=0	LI=1	WE=0	HALT=0	OUT=0
T=200000	PC=3	IR=40	AC=195	B=64	ALU=3	MEM_OUT=64	ADDR=3	T=000010	LB=0	LA=0	LI=0	WE=0	HALT=	

LOGISIM 8 BIT

Design Simul...

ARM1

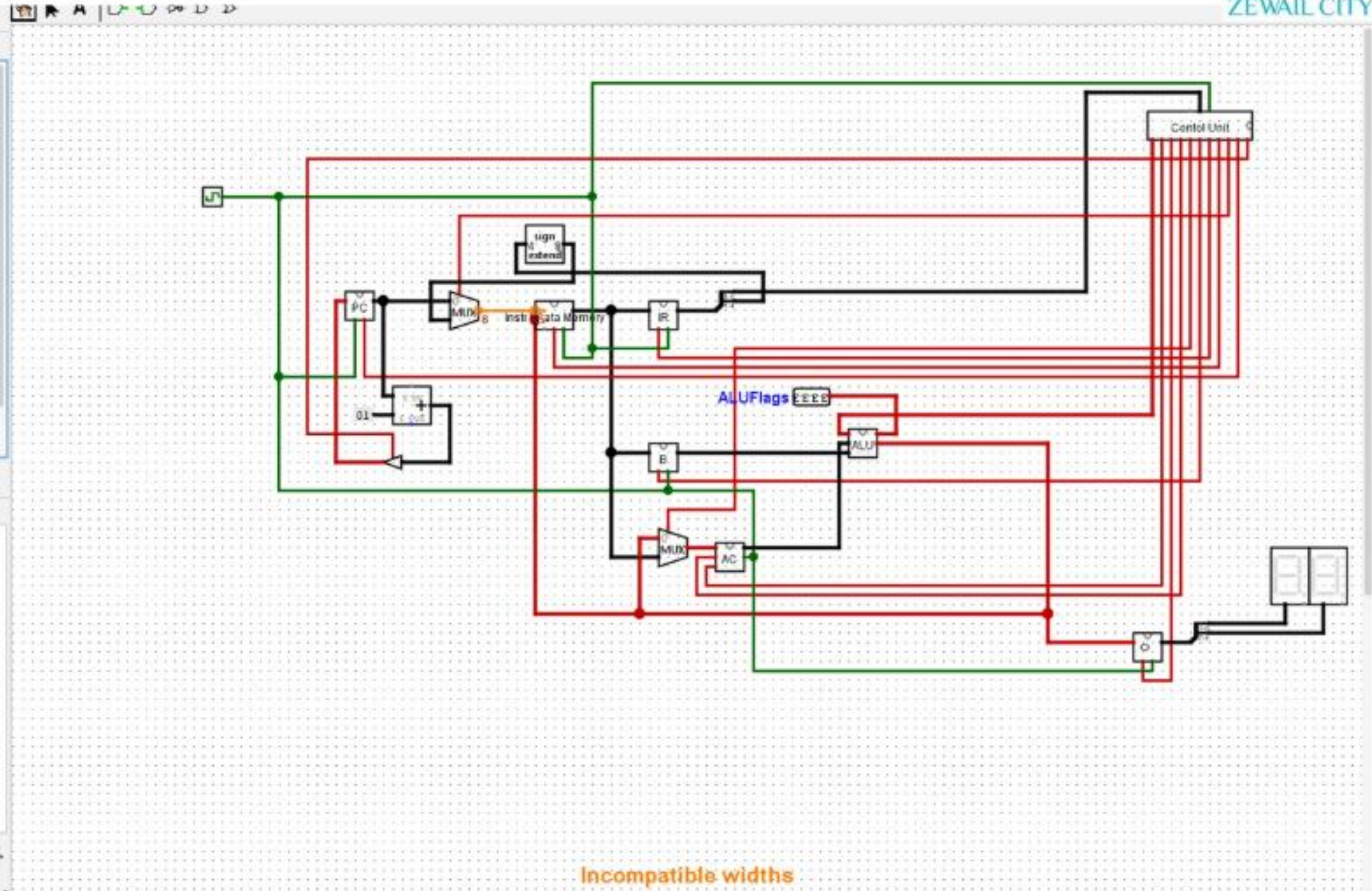
- main
- PC
- IR
- Instr_Data_Memory_e81c88ac
- AC
- B
- ALU
- O
- Control_Unit_bfab31f7
- reg
- Wiring
- Gates
- Plexers
- Arithmetic

Properties State

Circuit: main		
FPGA supported		Unknown
Circuit Name	main	
Shared Label		
Shared Label Facing	← West	
Shared Label Font	SansSerif Plain 12	
Appearance	Classic Logisim	
Use fixed box-size	No	

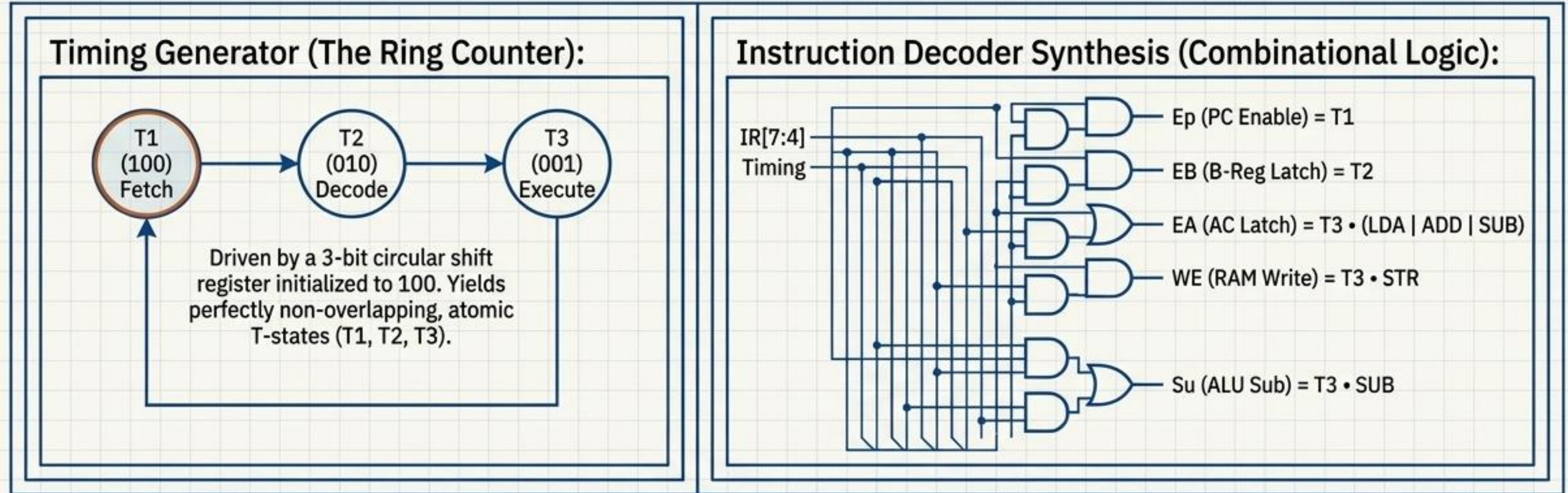
x0.8

Auto x1/3 x1 x2



Incompatible widths

Hardwired Control Unit: Zero Microcode Synthesized FSM



Timing Generator

Driven by a highly deterministic 3-bit circular shift register initialized to 100. Yields perfectly non-overlapping, atomic T-states (T1, T2, T3).

Decoder Synthesis Matrix

Relies entirely on combinational physical AND/OR gates. Avoids microcode ROM bloat to prioritize maximum efficiency and instant signal assertion.

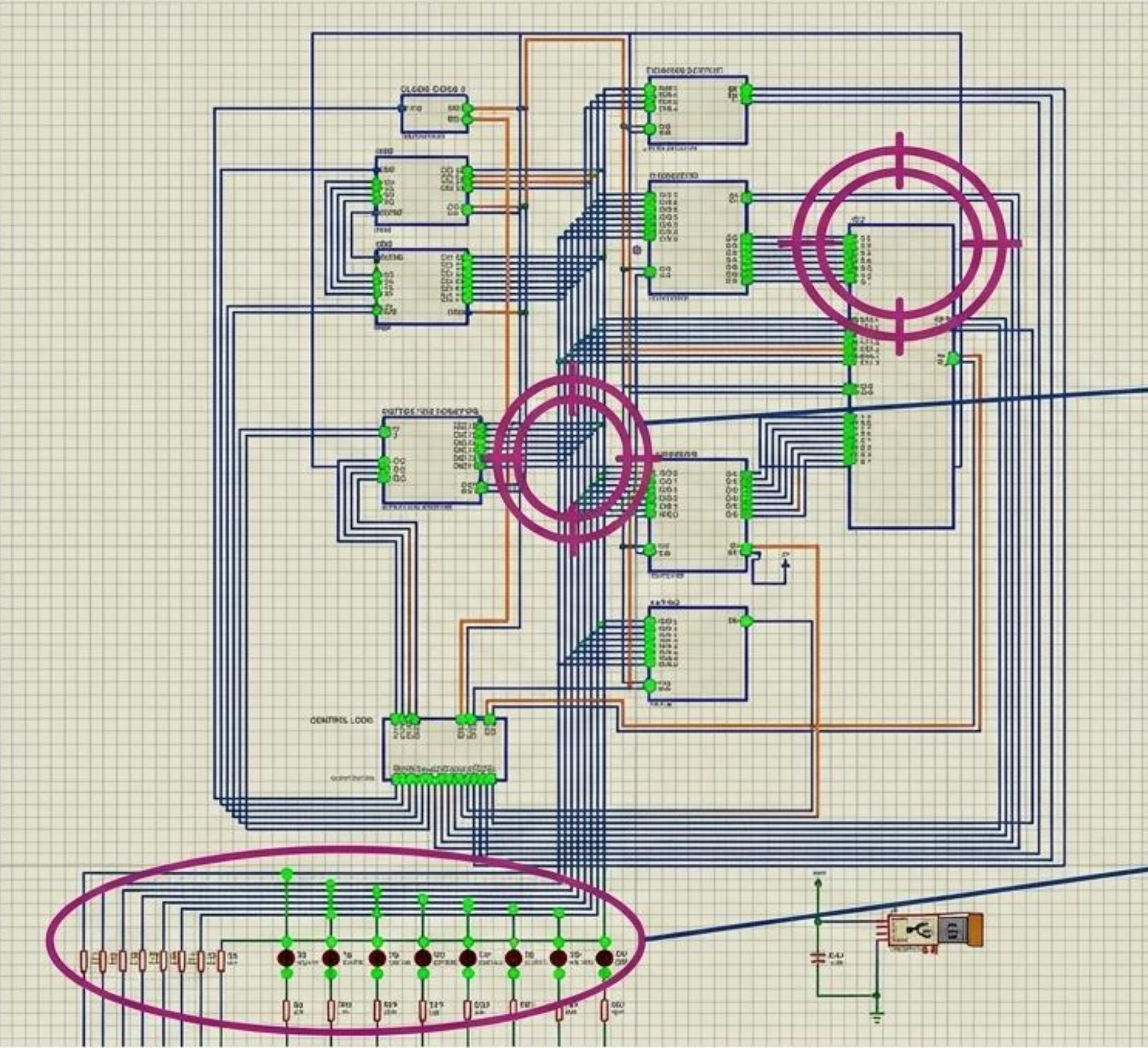
The HLT Trap

The Halt command physically cuts the system clock trace via a logical AND gate, freezing all registers indefinitely without utilizing complex branching logic.

ISA Specification & RTL Hardware Trigger Mapping

Opcode (Bin)	Mnemonic	Register Transfer Notation (RTL)	Hardware Trigger Description
0000	LDA addr	$AC \leftarrow M[IR[3:0]]$	Asserts EA; Ressel MUX routes RD bus to AC input.
0001	ADD addr	$AC \leftarrow AC + M[IR[3:0]]$	Asserts EA; Ressel routes ALU; ALU Su=0 (Addition).
0010	SUB addr	$AC \leftarrow AC + \sim M[IR[3:0]] + 1$	Asserts EA; Ressel routes ALU; ALU Su=1 (2's Comp Sub).
0101	STR addr	$M[IR[3:0]] \leftarrow AC$	Asserts WE; AC output drives WD bus; writes on inverted clock.
1110	OUT	$O_Reg \leftarrow AC$	Asserts EO; Output register latches AC bus to 7-segment display.
1111	HLT	$CLK_EN \leftarrow 0$	Triggers trap logic; permanently gates CLK signal to Ring Counter.

System-Level Digital Verification Strategy



Verification Checkpoints



Logisim FSM Verification

Technical Description:
Validated gate-level state transition matrices and non-overlapping control signal assertions prior to physical trace mapping.



Proteus Block Architecture

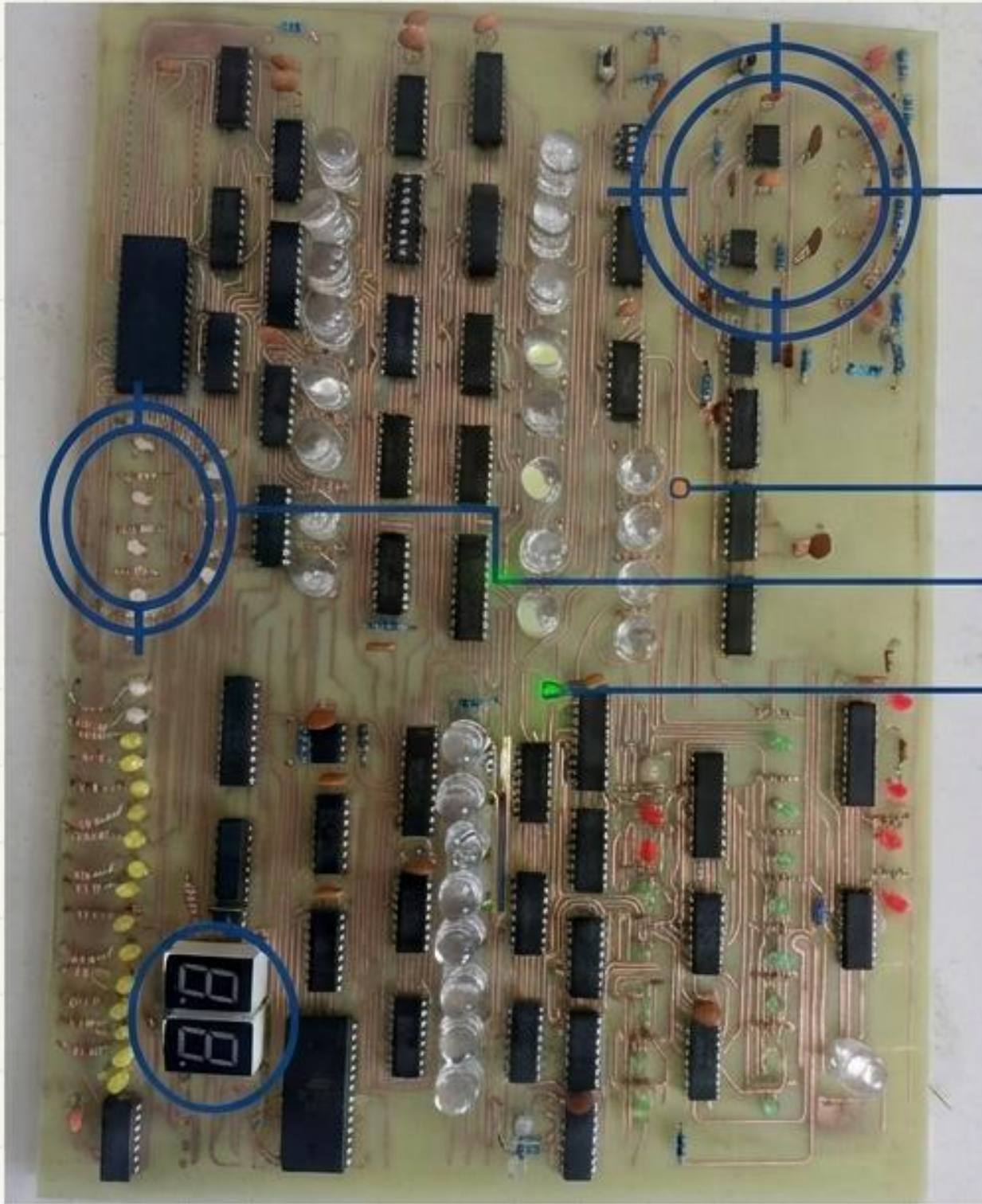
Technical Milestones:
Mapped abstractions to real-world 74-series discrete component models (MAR, RAM, Registers, ALU matrices).



Power & Diagnostic Emulation

Validated multiplexer propagation delays and visual state verification by tapping an 8-bit LED array directly into the Output register.

Discrete Logic Realization on Silicon



Pure Hardware Footprint

Fabricated entirely from standard 74-series TTL logic ICs. Zero embedded microcontrollers or pre-packaged CPU cores.

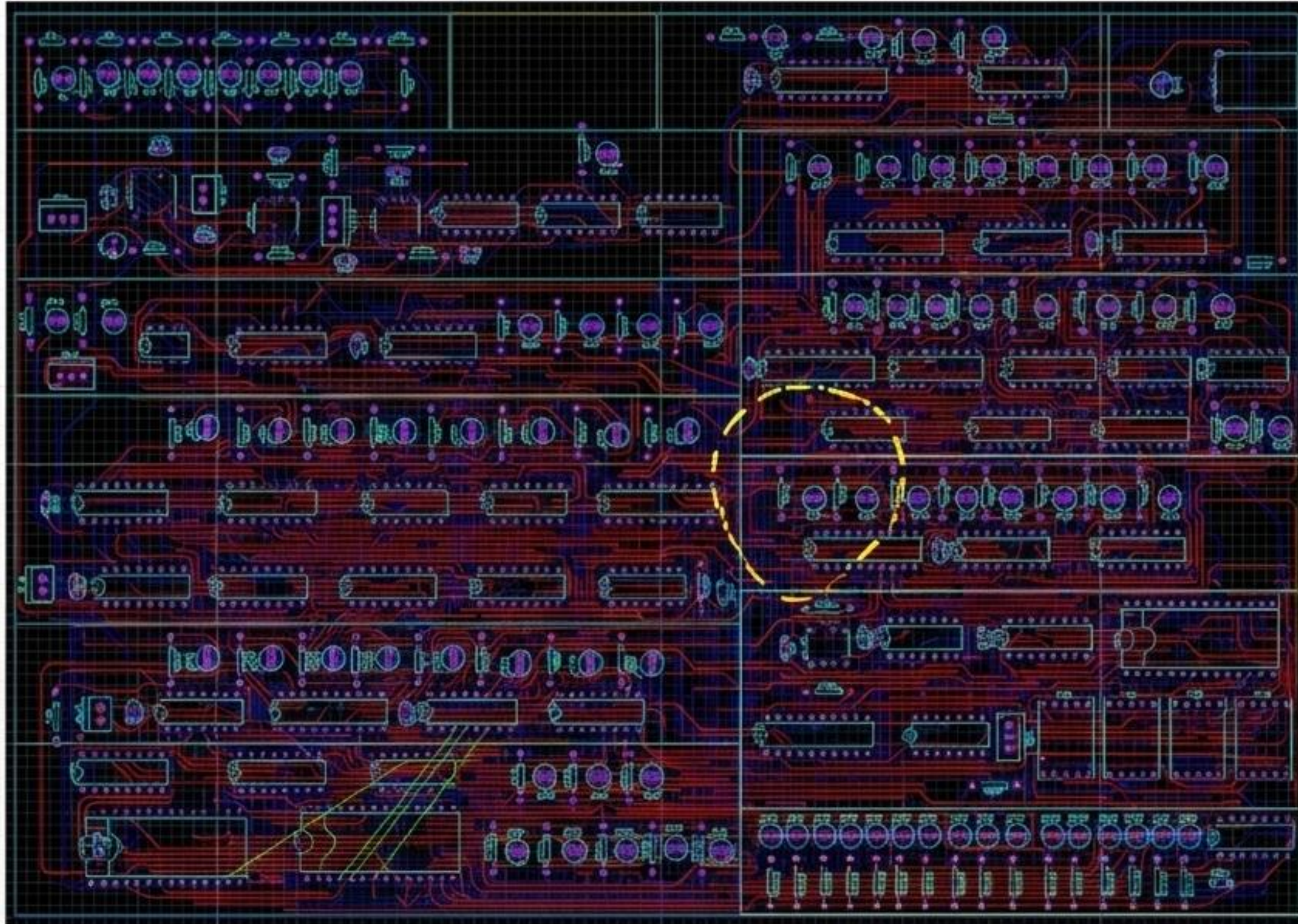
Manual Diagnostics

Hardware-debounced 555-timer circuit allows engineers to physically single-step through T1-T2-T3 execution cycles for deep debugging.

Real-Time Bus Monitoring

Vertical arrays of LEDs tapped directly onto data, address, and control buses acting as hardware-level logic probes.

High-Density Routing & Signal Integrity Management



Dual-Layer Isolation

Strict isolation of 8-bit parallel buses managed by separating horizontal (top copper/red) and vertical (bottom copper/blue) layers.

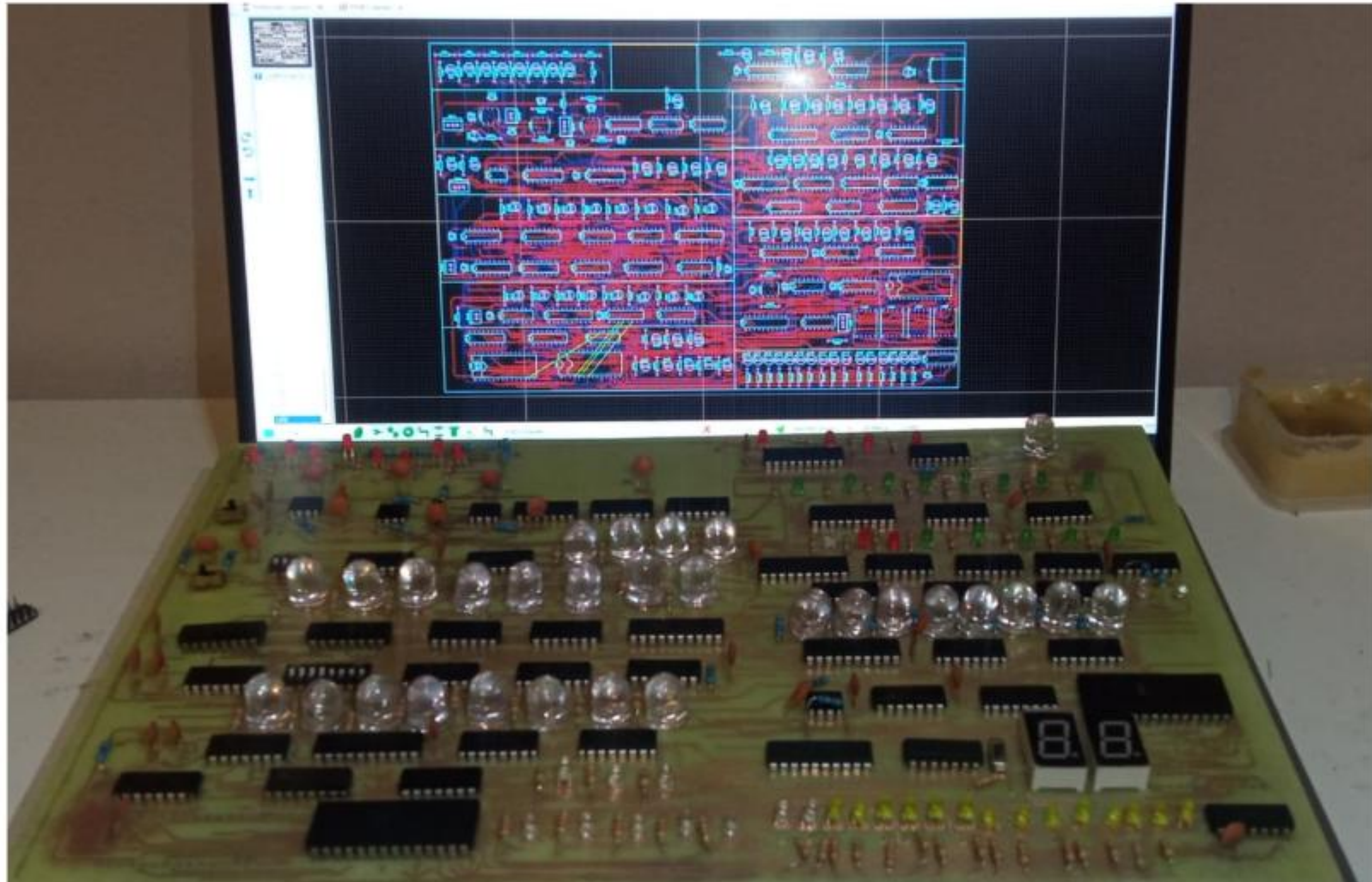
Component Grouping

ALU inputs routed physically adjacent to A and B register ICs to drastically minimize signal propagation delay across FR4 substrate.

Power Distribution Network (PDN)

Strategic bypass capacitors placed near IC power pins. Filters high-frequency switching noise, preventing cross-talk and false-triggering on CLK lines.

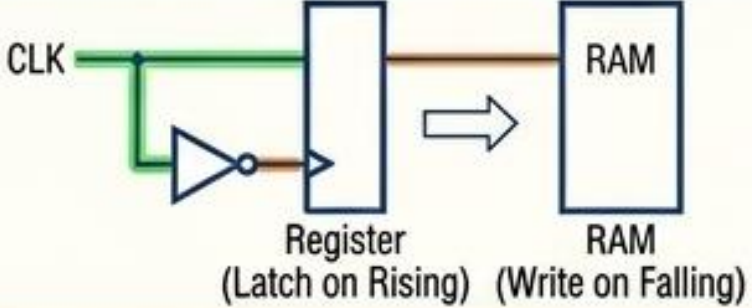
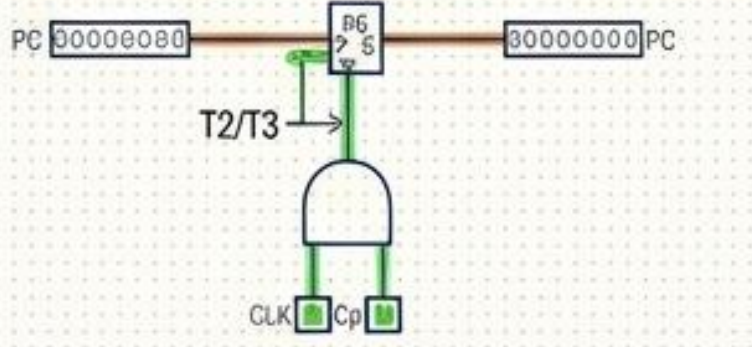
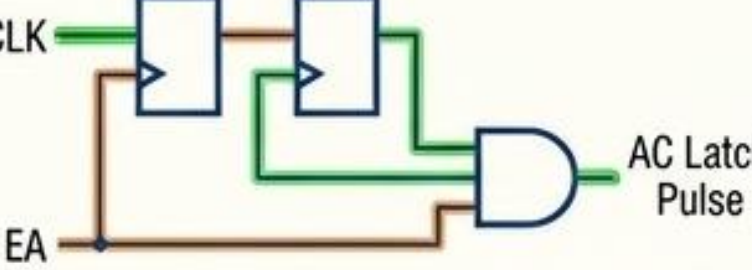
PCB 8 BIT







Hazard Mitigation: Resolving System Race Conditions

Hazard	Root Cause	Hardware Solution
Hazard 1: Write-Before-Stable	Rising edge simultaneously triggered register capture and RAM writes.	Inverted RAM clock; registers latch on rising edge, RAM securely writes on stable falling edge. 
Hazard 2: PC Bus Contention	PC adder inherently updated while frozen, risking bus shorts.	Switchable Controlled Buffer (Tri-state) physically disconnects adder during T2/T3 phases. 
Hazard 3: Accumulator Multi-Latching	Long duration Execute signals caused continuous AC re-latching.	Custom AND-gate pulse detector ensuring AC latches exactly once per Enable assertion. 

Architectural Limitations & The v2.0 Roadmap

Current System Limitations



Execution Limitation

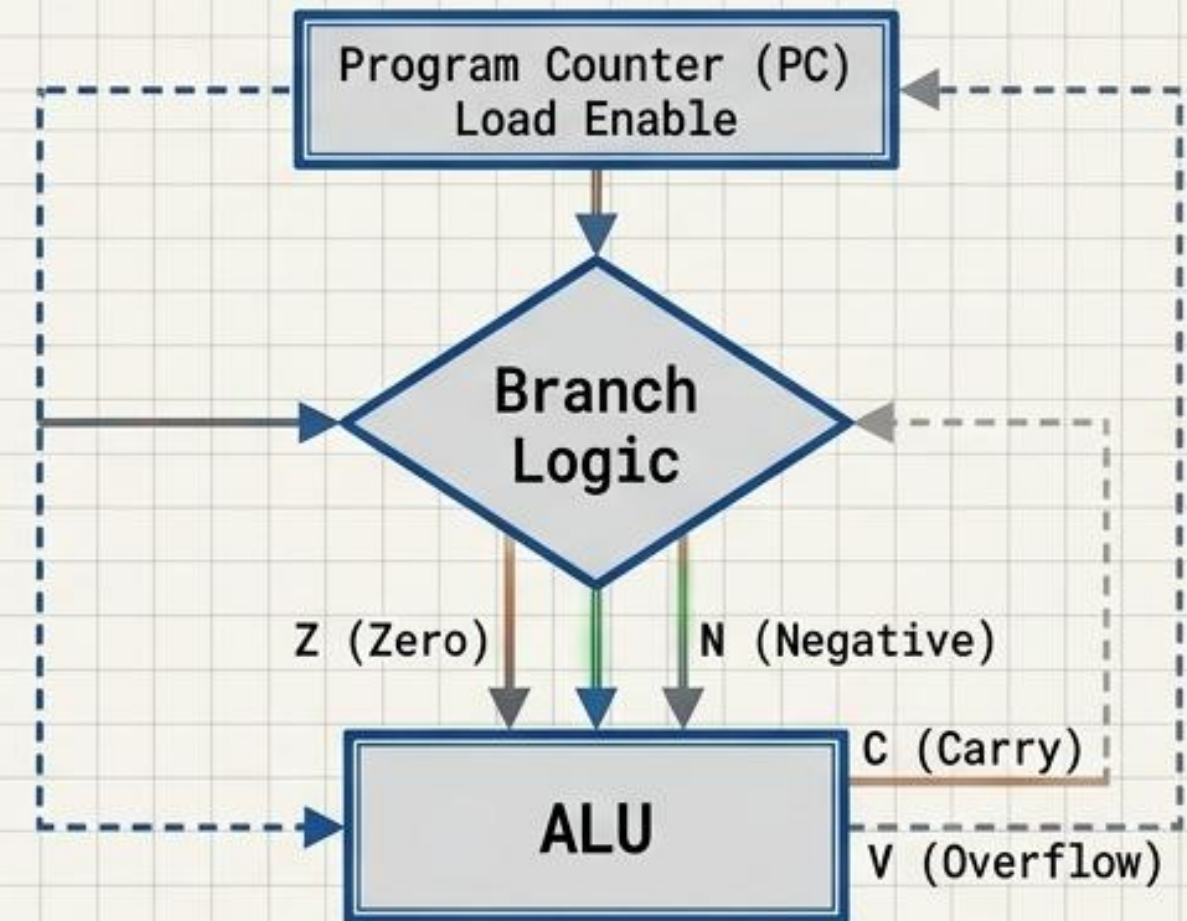
Processor currently executes linearly until a HLT command. Lacks conditional branching logic; not Turing complete.



Memory Constraint

4-bit address space restricts machine to 16 bytes. Requires a dual-word fetch mechanism to decouple instruction width from memory depth.

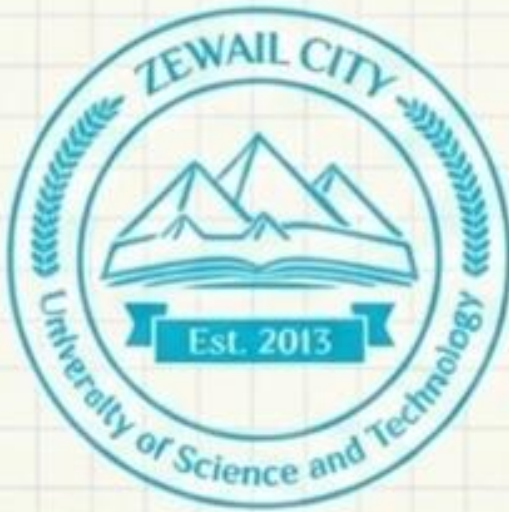
Proposed v2.0 Architecture Upgrades



Upgrade Path (Branch Logic):

Designing architecture for JZ (Jump if Zero) and JC (Jump if Carry). Routing ALU condition flags (Z, N, C, V) directly back into Program Counter's load-enable logic.

Project Conclusion & Open Defense



Core Competencies Demonstrated

Mastered hardwired FSM design, shared data bus multiplexing, high-density trace routing, and hardware race-condition mitigation.

Project Roster (Team 1)

Authored by the 20-member Sophomore Cohort including Moaz Elabd, Anas Refaei, Omar Ihab Fared, Ahmed Hany, Ahmed Ghoneim, Ahmed Akef, Mazen Adel, Abdallah Hosaam, Karim Assem, Mohamed Tamer, et al.

References

ARM Architecture foundational specs; Core principles of Multi-cycle Microarchitecture; Zewail City assignment guidelines.

